In [38]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import os
```

In [ ]:

In [ ]:

In [46]:
```
new_names=[
'word_freq_make',
'word_freq_address',
'word_freq_all',
'word_freq_3d',
'word_freq_our',
'word_freq_over',
'word_freq_remove',
'word_freq_internet',
'word_freq_order',
'word_freq_mail',
'word_freq_receive',
'word_freq_will',
'word_freq_people',
'word_freq_report',
'word_freq_addresses',
'word_freq_free',
'word_freq_business',
'word_freq_email',
'word_freq_you',
'word_freq_credit',
'word_freq_your',
'word_freq_font',
'word_freq_000',
'word_freq_money',
'word_freq_hp'    ,
'word_freq_hpl'    ,
'word_freq_george'  ,
'word_freq_650'     ,
'word_freq_lab'     ,
'word_freq_labs'    ,
'word_freq_telnet'  ,
'word_freq_857'     ,
'word_freq_data'    ,
'word_freq_415'     ,
'word_freq_85'      ,
'word_freq_technology',
'word_freq_1999'    ,
'word_freq_parts'   ,
'word_freq_pm'      ,
'word_freq_direct'  ,
'word_freq_cs'      ,
'word_freq_meeting' ,
```

```
'word_freq_original',
'word_freq_project' ,
'word_freq_re'      ,
'word_freq_edu'     ,
'word_freq_table'   ,
'word_freq_conference',
'char_freq_;'       ,
'char_freq_('       ,
'char_freq_['       ,
'char_freq_!'       ,
'char_freq_$'       ,
'char_freq_#'       ,
'capital_run_length_aver',
'capital_run_length_long',
'capital_run_length_total',
'spam'
]
```

In [47]:
```
df=pd.read_csv('spam.csv',header=None,names=new_names)
df.head()
```

Out[47]:

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d	word_freq_our	word_
0	0.00	0.64	0.64	0.0	0.32	
1	0.21	0.28	0.50	0.0	0.14	
2	0.06	0.00	0.71	0.0	1.23	
3	0.00	0.00	0.00	0.0	0.63	
4	0.00	0.00	0.00	0.0	0.63	

5 rows × 58 columns

In [52]:
```
print(df.isnull().sum().sum())
```

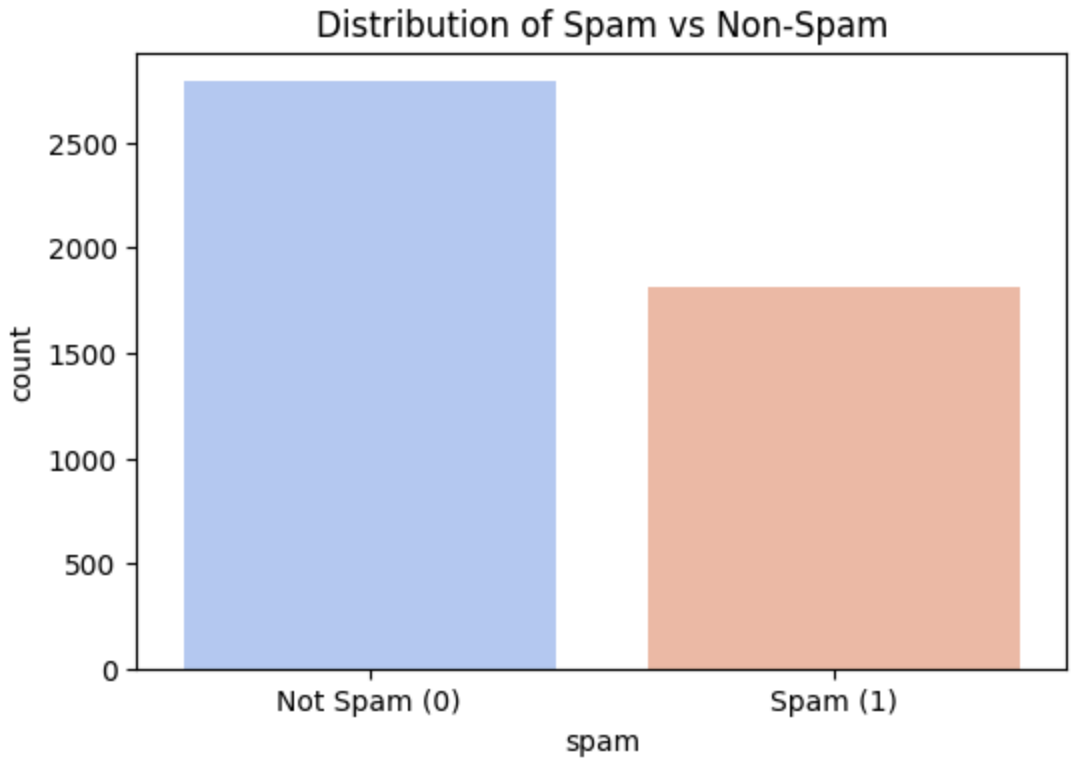
```
0
```

In [55]:
```
print("\nSpam Counts (1 = Spam, 0 = Not Spam):")
print(df['spam'].value_counts())
```

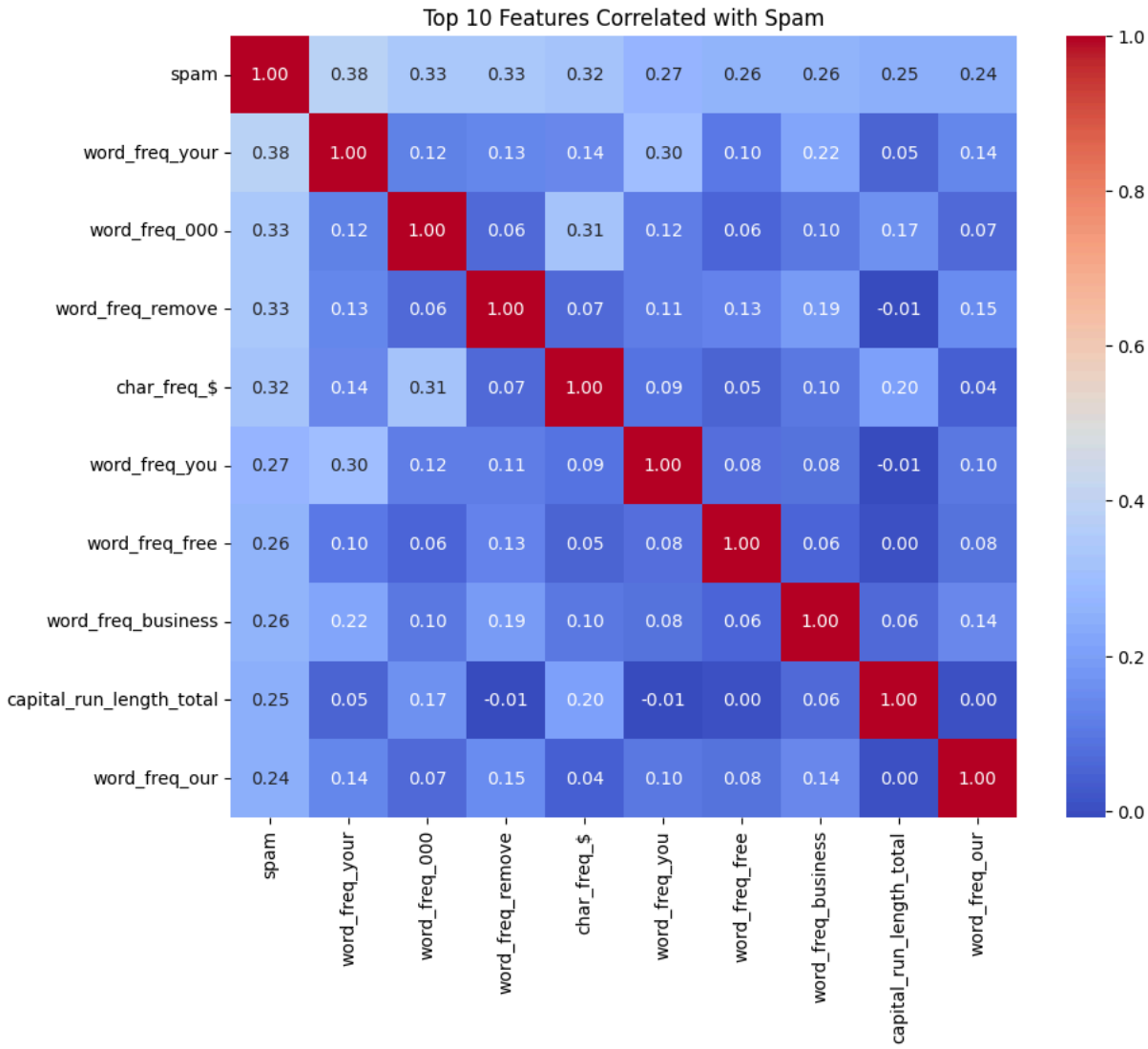
```
Spam Counts (1 = Spam, 0 = Not Spam):
spam
0    2788
1    1813
Name: count, dtype: int64
```

In [57]:
```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='spam', palette='coolwarm')
plt.title('Distribution of Spam vs Non-Spam')
plt.xticks([0, 1], ['Not Spam (0)', 'Spam (1)'])
plt.show()
```

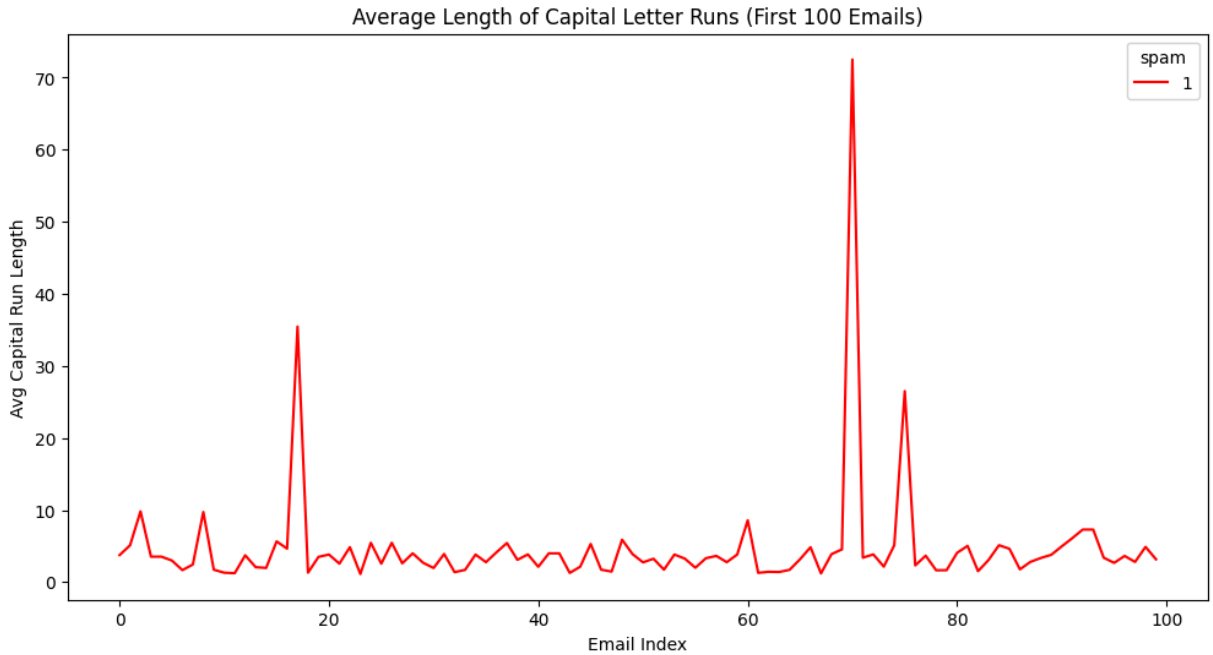
```
C:\Users\krish\AppData\Local\Temp\ipykernel_17428\2205555329.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(data=df, x='spam', palette='coolwarm')
```



In [61]:
```
plt.figure(figsize=(10,8))
corr_matrix=df.corr()
k=10
top_features=corr_matrix.nlargest(k,'spam')['spam'].index
sns.heatmap(df[top_features].corr(),annot=True,cmap='coolwarm',fmt='.2f')
plt.title('Top 10 Features Correlated with Spam')
plt.show()
```

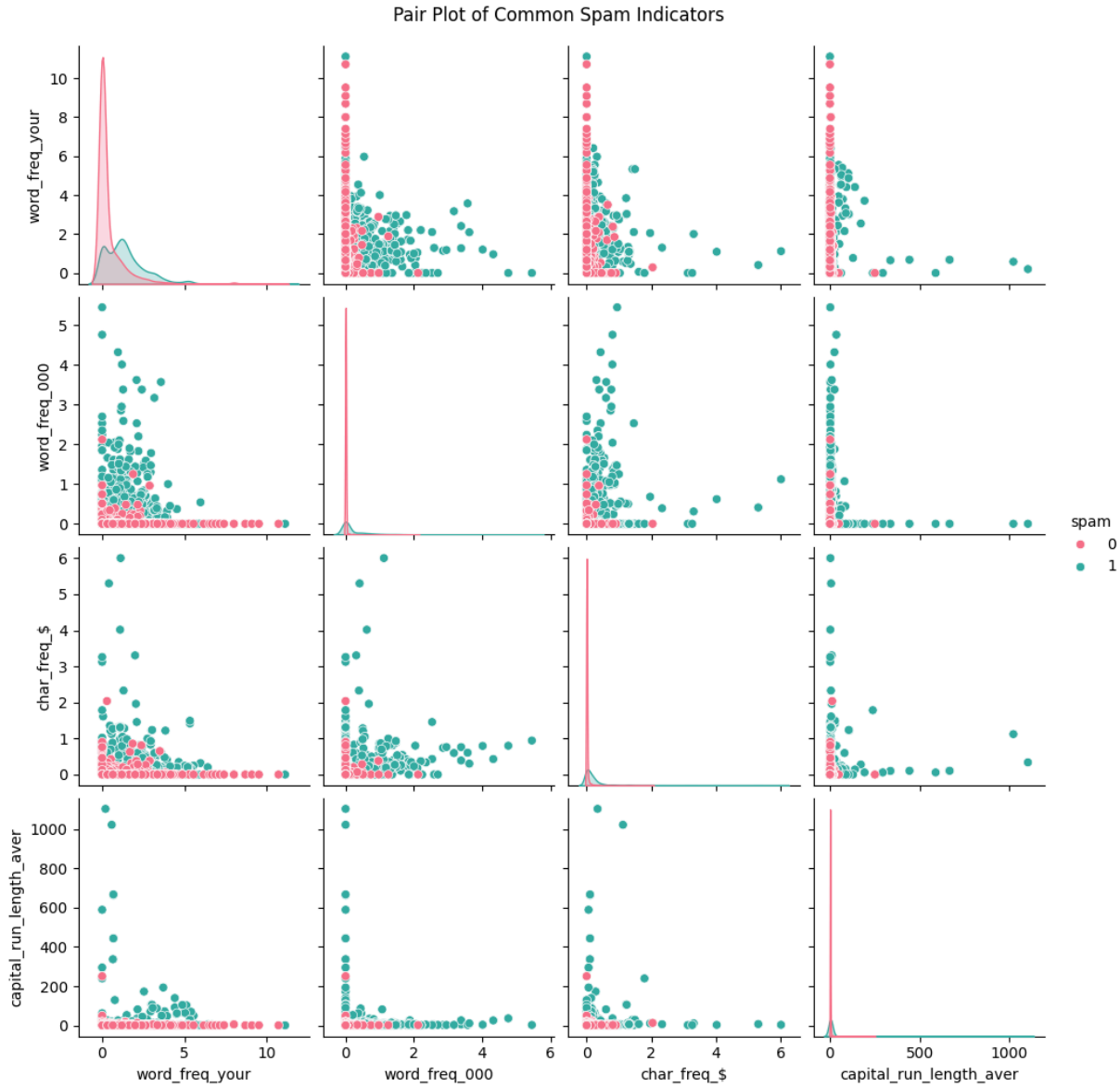


In [64]:
```
plt.figure(figsize=(12, 6))

subset = df.iloc[:100]

sns.lineplot(data=subset, x=subset.index, y='capital_run_length_aver', hue='spam',

plt.title('Average Length of Capital Letter Runs (First 100 Emails)')
plt.xlabel('Email Index')
plt.ylabel('Avg Capital Run Length')
plt.show()
```



In [66]:
```
selected_cols = ['word_freq_your', 'word_freq_000', 'char_freq_$', 'capital_run_len

plt.figure(figsize=(10, 8))
sns.pairplot(df[selected_cols], hue='spam', diag_kind='kde', palette='husl')
plt.suptitle('Pair Plot of Common Spam Indicators', y=1.02)
plt.show()
```

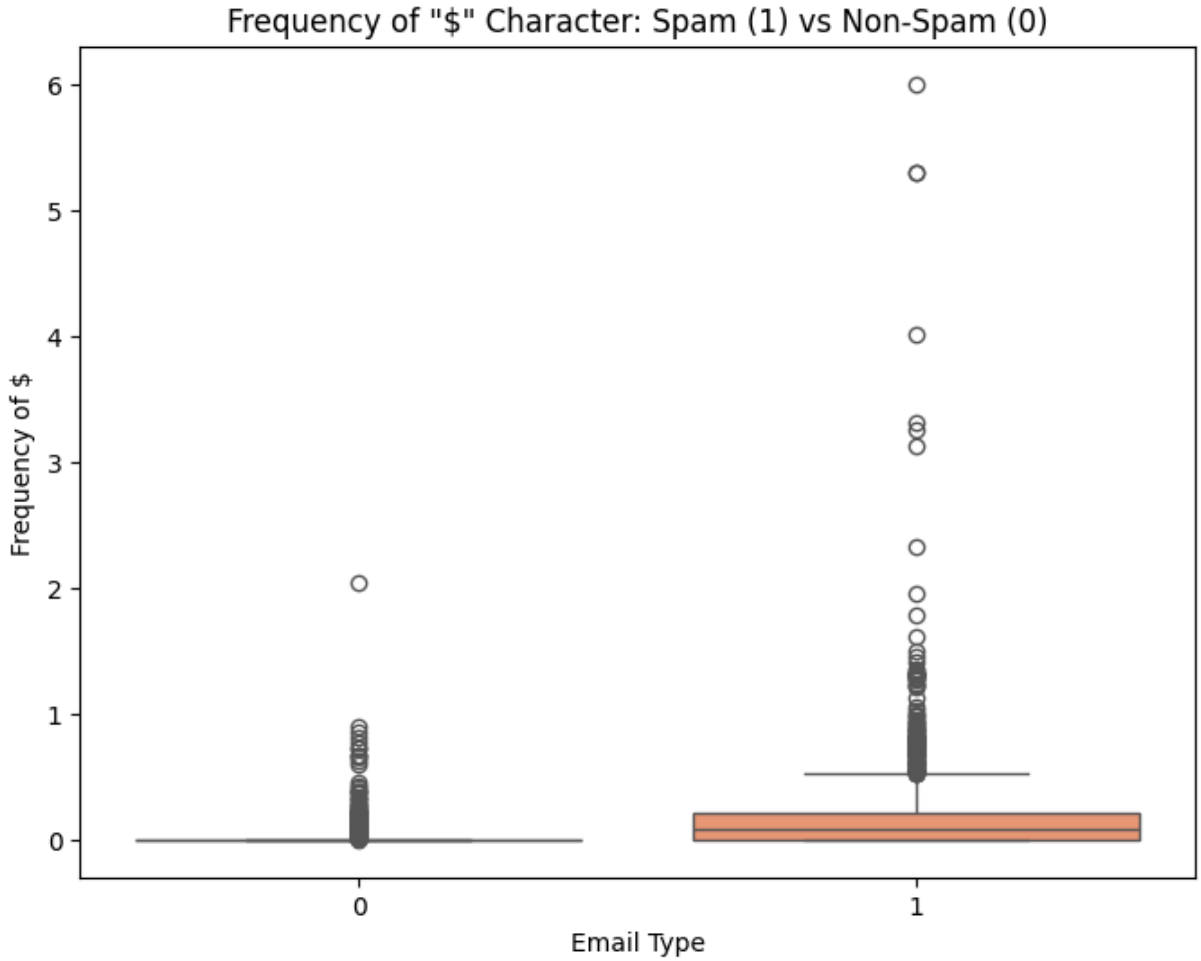
```
<Figure size 1000x800 with 0 Axes>
```



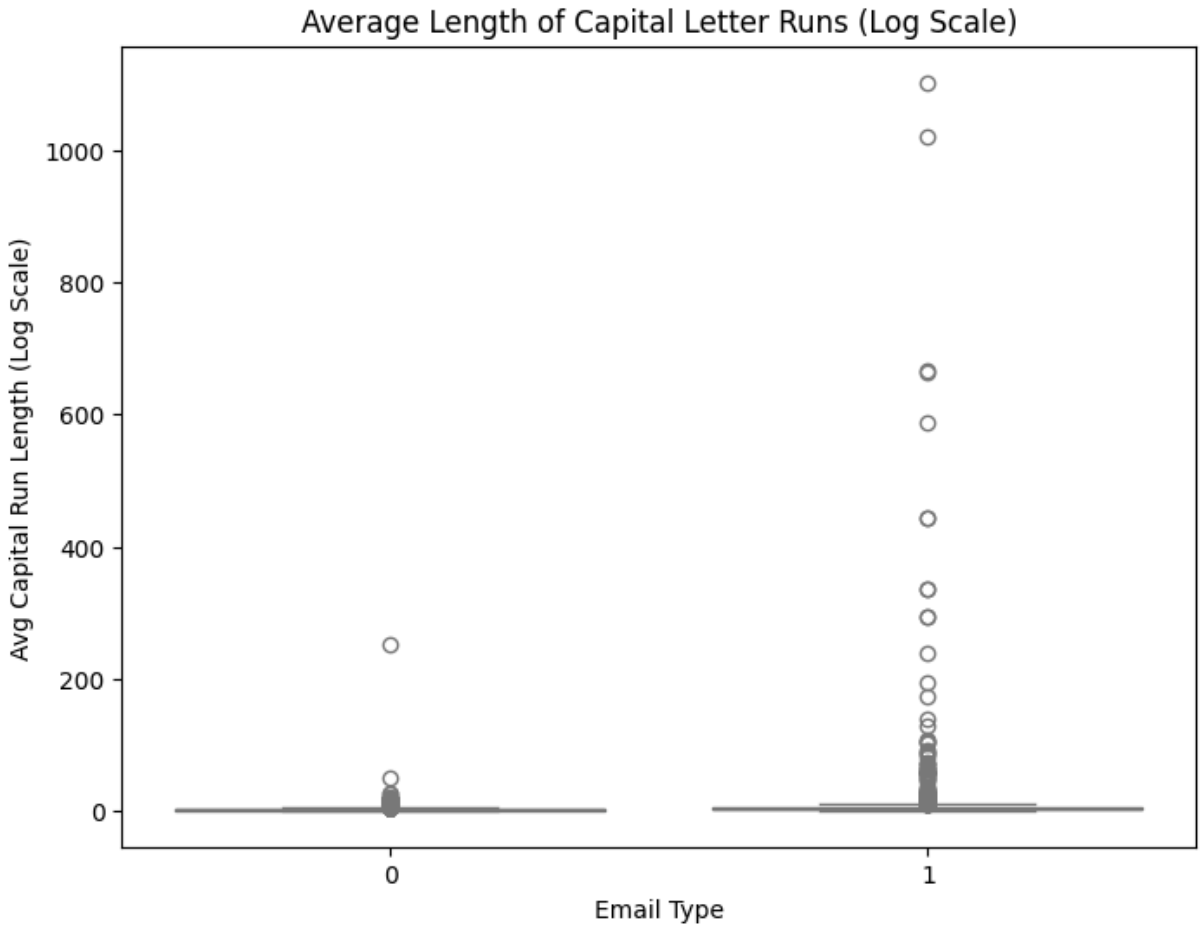
In [71]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Comparison of Dollar Sign Frequency ($)
plt.figure(figsize=(8, 6))

# We use 'sym="""' to hide the thousands of outlier points if you want a cleaner loo
# but usually, seeing the outliers is the point!
sns.boxplot(data=df, x='spam', y='char_freq_$', palette='Set2')

plt.title('Frequency of "$" Character: Spam (1) vs Non-Spam (0)')
plt.xlabel('Email Type')
plt.ylabel('Frequency of $')
plt.show()

# 2. Comparison of Capital Run Length (The "SHOUTING" metric)
# Since outliers are huge here, we use a Log Scale to make it readable
plt.figure(figsize=(8, 6))
sns.boxplot(data=df, x='spam', y='capital_run_length_aver', palette='coolwarm')
```

```
plt.title('Average Length of Capital Letter Runs (Log Scale)')
plt.xlabel('Email Type')
plt.ylabel('Avg Capital Run Length (Log Scale)')
plt.show()
```

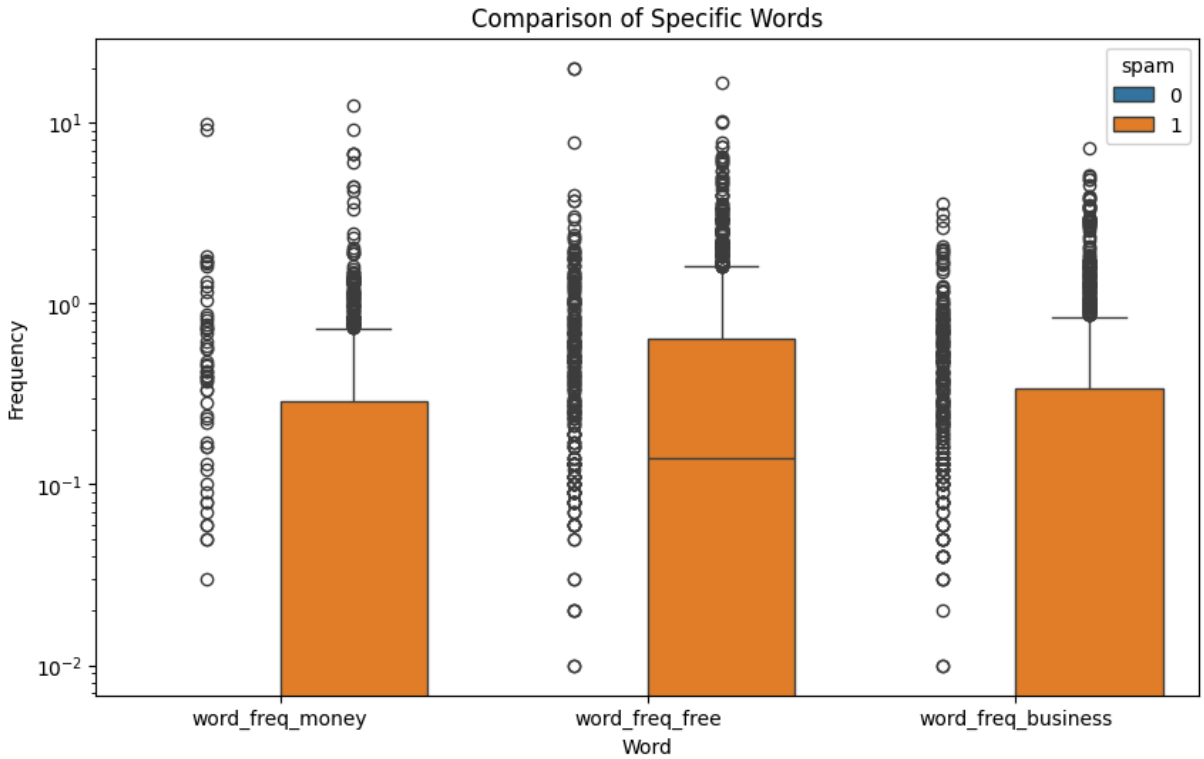
```
C:\Users\krish\AppData\Local\Temp\ipykernel_17428\824902335.py:10: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(data=df, x='spam', y='char_freq_$', palette='Set2')
```




```
C:\Users\krish\AppData\Local\Temp\ipykernel_17428\824902335.py:20: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(data=df, x='spam', y='capital_run_length_aver', palette='coolwarm')
```



In [73]:
```
features_to_plot = ['word_freq_money', 'word_freq_free', 'word_freq_business']


df_melted = df.melt(id_vars='spam', value_vars=features_to_plot, var_name='Word', v

plt.figure(figsize=(10, 6))
sns.boxplot(data=df_melted, x='Word', y='Frequency', hue='spam')
plt.title('Comparison of Specific Words')
plt.yscale('log')
plt.show()
```


Comparison of Specific Words
spam
0
1
Frequency
10^1
10^0
10^{-1}
10^{-2}
word_freq_money
word_freq_free
word_freq_business
Word

In [ ]: